

1. Which one of the following is correct about dissolved oxygen (DO) concentration in water sources?
 - A. High DO concentration is correlated with high productivity and little pollution.
 - B. Low DO concentration indicates no water pollution.
 - C. High DO concentration indicates high biological oxygen demand (BOD)
 - D. The DO concentration does not indicate the degree of water pollution.
2. What is the five-day and seven-day BOD for a sample of treated wastewater assuming a BOD rate constant, $k = 0.10 \text{ days}^{-1}$ and an ultimate BOD of 350 mg/L?
 - A. 122mg/l and 161 mg/l
 - B. 220 mg/l and 240 mg/l
 - C. 138 mg/L and 176mg/L
 - D. 178 mg/l and 205 mg/l
3. Which one of the following is the most preferable and recommended waste management hierarchical order?
 - A. Recycling, Reuse, Incineration, Land fill
 - B. Land fill, Incineration, Reuse, Recycling,
 - C. Recycle, Reuse, Incineration, Land fill
 - D. Reuse, Recycling, incineration, Land fill
4. Taking into account cost, ease of operation, and safety, what is the best disposal options for the following solid waste types?
 - I. Cellulosic, paper and plastic wastes
 - II. Biodegradable or putrescible wastes
 - III. Ashes released from power plants
 - IV. Sludge from wastewater treatment plant
 - A. (I) Landfills, (II) Land fill, (III) Combustion, (IV) Composting and digestion
 - B. (I) Combustion, (II) Composting and digestion, (II) Landfills, (IV) Land fill
 - C. (I) Composting and digestion (II) Landfills (III)Land fill (IV) Combustion
 - D. (I) Landfills, (II) Composting and digestion, (III)Land fill, (IV) Combustion
5. One of the following meteorological elements has no significant effects on the distribution of air pollutants.

- A. Wind speed and direction
 - B. Solar radiation
 - C. Stability and precipitation
 - D. Light intensity
6. ----- uses centrifugal forces to separate particulate matter from air pollutant streams.
- A. Cyclone
 - B. Electrostatic precipitators
 - C. Fabric filter
 - D. Adsorbed
7. Which of the following technique **is not** possibly used to reduce tube-side pressure drop during heat exchanger design?
- A. Reducing tube-side velocity
 - B. Reducing baffle spacing
 - C. Reducing heat transfer coefficient
 - D. Increasing number of tubes per pass
8. Which of the following heads and closures can be used for high pressures operating processes?
- A. Flange-only/Flat plates
 - B. Torispherical
 - C. Ellipsoidal
 - D. Hemispherical
9. Which of the following support is used for the horizontal pressure vessels?
- a. Leg support
 - b. Saddle support
 - c. Skirt support
 - d. Lug support
10. In evaporator design if the concentrated product may degenerate if exposed to high temperature, which feeding arrangement is best?
- A. Mixed feed
 - B. Backward feed
 - C. Forward feed
 - D. Parallel feed
11. If maximum in pressure vessel is 3.5N/mm^2 , diameter=200mm and permissible tensile stress is 40N/mm^2 , calculate thickness of cylinder wall considering corrosion allowance:
- A. 10.75mm
 - B. 8.75mm
 - C. 9.75mm
 - D. 12.75mm
12. A competitive inhibitor of an enzyme is usually
- A. a highly reactive compound
 - B. a metal ion such as Hg^{2+} or Pb^{2+}
 - C. structurally similar to the substrate.

D. water-insoluble

13. The phenomenon in which substrates are used sequentially is known as

- A. Dialism
- B. Trans-substrate genesis
- C. Diauxie
- D. Multiplicity

14. The dilution rate, D is defined as (where F = volumetric flow rate, V_R = total volume of culture in the reactor and μ specific growth rate)

- A. F/V_R
- B. V_R/F
- C. μ/F
- D. F/μ

15. An open system in which the growth rate is maintained by the removal and addition of media at such a rate as to maintain a constant cell density is called a

- A. Manostat
- B. Chemostat
- C. Turbidostat
- D. Culturostat

16. . The moist heat is more effective than the dry heat because the intrinsic heat resistance of vegetative cells is greatly

- A. increased in a dry state
- B. decreased in a dry state
- C. increased in a wet state
- D. decreased in a wet state

17. Typical aeration rates for aerobic fermentation are

- A. 0 - 0.5 vvm
- B. 0.5 - 1.0 vvm
- C. 1.0 - 1.5 vvm
- D. 1.5 - 2.0 vvm

18. What causes cavitation in centrifugal pump?

- A. High suction pressure
- B. Low barometric pressure
- C. Low suction pressure
- D. High suction velocity

19. What feature distinguish reciprocating pumps compared to centrifugal pumps

- A. Deliver liquid at uniform pressure.

- B. Can handle slurries more efficiently.
 - C. Are not subject to air binding.
 - D. Can be operated with delivery valve closed.
20. Pumping of a corrosive liquid is generally preferred to be done by a _____ pump, as it can be made of a variety of materials including plastics.
- A. piston
 - B. gear
 - C. positive displacement
 - D. sump
21. The excess of the sum of pressure & velocity heads over the vapor pressure of the liquid at the suction is called the
- A. static submergence.
 - B. net positive suction head (NPSH).
 - C. cavitation sensitivity.
 - D. priming.
22. The energy loss over a length of pipeline according to Darcy-Weisbach equation for pipe flow is.....the mean velocity of flow.
- A. directly proportional to
 - B. directly proportional to square of
 - C. inversely proportional to
 - D. inversely proportional to square of
23. Which of the following can be used to create a flow of gas, where no significant compression is required?
- A. Reciprocating compressor
 - B. Blower
 - C. Axial flow compressor
 - D. Centrifugal compressor
24. Two fluids 1 and 2 have mass densities of ρ_1 and ρ_2 respectively. If $\rho_1 > \rho_2$, which one of the following expressions will represent the relation between their specific volumes v_1 and v_2 ?
- A. $v_1 > v_2$
 - B. $v_1 < v_2$
 - C. $v_1 = v_2$
25. The rise in the level of a liquid in a tube is h . What will be the rise in the level if the same amount of liquid is poured into a tube of half the diameter?

- A. 0
- B. $h/2$
- C. h
- D. $2h$

26. Which property of the fluid accounts for the major losses in pipes?

- A. density
- B. specific gravity
- C. viscosity
- D. compressibility

27. A soap bubble of d mm diameter is observed inside a bucket of water. If the pressure inside the bubble is 0.075 N/cm^2 , what will be the value of d ? (Take surface tension as 0.075 N/m)

- A. 0.4
- B. b) 0.8
- C. 1.6
- D. 4

28. When is a fluid flow called turbulent?

- A. High viscosity of fluid
- B. Reynolds number is greater than 2000
- C. Reynolds number is less than 2000
- D. The density of the fluid is low.

29. Which device is popularly used for measuring difference of low pressure?

- a) Inverted U-tube Differential Manometer
- b) U-tube Differential Manometer
- c) Inclined Single column manometer
- d) Vertical Single column manometer

30. It is required to prepare 1250 kg of a solution composed of 12 wt.% ethanol and 88 wt.% water. Two solutions (A and B) are available, the first (A) contains 5 wt.% ethanol, and the second (B) contains 25 wt.% ethanol. How much of each solution are mixed to prepare the desired solution?

- A. 803.5 kg of A and 446.5 kg of B
- B. 812.5 kg of A and 437.5 kg of B
- C. 907.5 kg of A and 342.5 kg of B
- D. 903.5 kg of A and 346.5 kg of B

31. A _____ stream is one that skips one or more stages of the process and goes directly to another downstream stage.
- Recycle
 - Purge
 - Bypass
 - None of the above
32. 20 moles of O_2 is added to 20 moles of H_2 , how many moles of H_2O will it produce?
- 5
 - 10
 - 15
 - 20
33. Among the following unit operations, find the odd one
- Condensation
 - Evaporation
 - Distillation
 - Adsorption
34. The core services of Chemical Engineer are not included
- Economic feasibility study
 - Construction, operation and maintenance of pilot plant
 - Establishing and conducting of quality control
 - Sewerage assessment
35. Which one of the following is a dimensionless quantity?
- Mass
 - Weight
 - Specific weight
 - Reynold's number
36. Coefficient of thermal conductivity is defined as the heat flow per unit time.
- when temperature gradient is unity
 - across unit area when temperature gradient is unity
 - across unit area
 - through unit thickness
37. Rate of heat conduction through a thick sphere of radius r_1 and r_2 is equal to.
- $\frac{4\pi k(t_1 - t_2)}{(r_2 - r_1)}$
 - $\frac{4\pi k r_1 r_2 (t_1 - t_2)}{(r_2 - r_1)}$

C. $\frac{4\pi k r_1 r_2}{(r_2 - r_1)}$

38. Ratio of emissive power of a body to the emissive power of a perfectly black body, is defined as
- emissivity
 - absorptivity
 - transmissivity
 - reflectivity
39. The overall heat transfer coefficient (U) for a composite wall of thickness x_1, x_2, x_3 , and of corresponding thermal conductivities k_1, k_2, k_3 is given by the equation (Analyzing)
- $1/U = k_1/x_1 + k_2/x_2 = k_3/x_3$
 - $U = k_1/x_1 + k_2/x_2 = k_3/x_3$
 - $1/U = x_1/k_1 + x_2/k_2 + x_3/k_3$
 - $U = x_1/k_1 + x_2/k_2 + x_3/k_3$
40. Transient three-dimensional heat conduction is governed by one of the following differential equations (α = thermal diffusivity, K = thermal conductivity and e = volumetric rate of heat generation)
- $1/\alpha * \frac{\partial T}{\partial t} = \nabla T + ek$
 - $1/\alpha * \frac{\partial T}{\partial t} = \nabla T + e/k$
 - $1/\alpha * \frac{\partial T}{\partial t} = \nabla^2 T + ek$
 - $1/\alpha * \frac{\partial T}{\partial t} = \nabla^2 T + e/k$
41. The Fourier number is used in the analysis of problem involving heat transfer by
- Forced convection
 - Natural convection
 - Transient conduction
 - Steady state conduction
42. The reason for preferring packed towers over plate towers in distillation practice is that the packed tower operation gives
- low pressure drop and high hold up.
 - high pressure drop and low hold up.
 - low pressure drop and low hold up.
 - high pressure drop and high hold up.
43. Find the time taken to dry 50% to 40% of moisture (dry basis) if critical moisture content is 30%. If the weight of the wet solid is 5 kg and the product of area and the constant rate is 0.5 kg/min.
- 20 sec

- B. 40 sec
 - C. 60 sec
 - D. 80 sec
44. Positive deviation from Raoult's law means a mixture whose total pressure is
- A. greater than that computed for ideality.
 - B. less than that computed for ideality.
 - C. less than the sum of the vapour pressure of the components.
 - D. none of these.
45. If the feed enters the distillation column at its dew point, then the slope of the Flash vaporization operating line is
- A. Infinity
 - B. 0
 - C. 1
 - D. Between 0 and 1
46. -----is preferred for the binary mixture having component A and B with vapour pressure 360mmHg and 355mmHg.
- A. Multi-component distillation
 - B. Reactive distillation
 - C. Azeotropic distillation
 - D. Fractionating distillation
47. In liquid extraction, if the selectivity is unity, then
- A. separation of the constituents is the most effective.
 - B. no separation will occur.
 - C. amount of solvent required will be minimum.
 - D. solvent flow rate should be very low
48. Component A is diffusing in a medium B. The flux N_A relative to a stationary point is equal to the flux due to molecular diffusion, if
- A. mass transfer is accompanied by reaction.
 - B. diffusion of A is in stagnant medium B.
 - C. molecular mean free path is high.
 - D. there is equimolar counter diffusion.
49. A piece of equipment originally costing \$40,000 was put into use 12 years ago. At the time the equipment was put into use, the service life was estimated to be 20 years and the salvage and scrap value at the end of the service life were assumed to be zero. On this basis, a straight-line

depreciation fund was set up. The equipment can now be sold for \$10,000, and a more advanced model can be installed for \$55,000. Assuming the depreciation fund is available for use, how much new capital must be supplied to make the purchase?

- A. \$35,000
- B. \$2,000
- C. \$21,000
- D. \$24,000

50. In the area of plant design and Engineering Economics, which one of the following definition describes the term “payback period”?

- A. the total amount of money needed to supply the necessary plant and manufacturing facilities
- B. The amount of money required as working capital for operation of the facilities.
- C. The length of time it takes to recover the cost of an investment
- D. Is the net gain or loss of an investment over a specified time period, expressed as a percentage of the investment's initial cost.

51. When considering possible ways of achieving the objective the designer will be constrained by many factors which are called the design constraints. Which one of the following is EXTERNAL constraint?

- A. Choice of Process
- B. Process Condition
- C. Safety Regulation
- D. Methods

52. Expenditure on research and development (R & D) is categorized as the _____, while making an estimate of the total product cost for a chemical plant.

- A. Overhead cost
- B. Fixed expenses
- C. General expenses
- D. Direct production cost

53. Which of the following is not a component of working capital?

- a. Raw materials is stock
- b. Finished products in stock
- c. Transportation facilities
- d. Semi-finished products in the process

54. In a manufacturing industry, breakeven point occurs, when the

- A. Total annual rate of production equals the assigned value
- B. Total annual product cost equals the total annual sales
- C. Annual profit equals the expected value
- D. Annual sales equals the fixed cost

55. Which of the following operation is not included in Beam house operation?

- A. Fleshing
- B. Soaking and washing
- C. Liming
- D. Trimming

56. Which of the following statement explains the advantages of using precalciner?

- A. Larger kiln size
- B. fuel efficiency
- C. sound cement
- D. fine grained cement

57. _____ is a method for producing hydrogen, carbon monoxide, or other useful products from hydrocarbon fuels such as natural gas

- A. carbonation
- B. Meth nation
- C. Steam reforming
- D. dehydration process

58. Which of the following nitrogenous fertilizer contains the highest nitrogen in percent?

- A. DAP
- B. MAP
- C. NH_3
- D. Ammonium sulfate

59. High concentration of NaOH is produced by _____

- A. Mercury cell process
- B. Diaphragm cell process
- C. Membrane cell process
- D. Electrolysis process

60. Which of the following chemical product belongs to sulfuric acid production process?

- A. $\text{H}_2\text{S}_2\text{O}_7$
- B. $\text{H}_2\text{S}_2\text{O}_8$
- C. $\text{H}_2\text{S}_2\text{O}_6$
- D. $\text{H}_2\text{S}_2\text{O}_5$

61. In a heat exchanger, floating head is provided to

- A. Facilitate cleaning of the exchanger.
- B. Increase the heat transfer area.
- C. Relieve stresses caused by thermal expansion.
- D. Increase log mean temperature gradient.

62. In the equation $Q = UA\Delta t$; Δt is
- A. Geometric mean temperature difference.
 - B. Arithmetic mean temperature difference.
 - C. Logarithmic mean temperature difference.
 - D. The difference of average bulk temperatures of hot and cold fluids.
63. In case of parallel flow heat exchanger, the lowest temperature theoretically attainable by the hot fluid is
- A. Equal to the outlet temperature of the cold fluid
 - B. More than the outlet temperature of the cold fluid
 - C. Less than the outlet temperature of the cold fluid
 - D. Either more or less than the outlet temperature of the cold fluid (depending upon the fluid)
64. In a liquid-liquid heat exchanger, for the same process temperature, the ratio of the LMTD in parallel flow to the LMTD in counter flow is always
- A. < 1
 - B. > 1
 - C. 1
 - D. ∞
65. In a heat exchanger it is observed that, $\Delta T_1 = \Delta T_2$, where ΔT_1 is the temperature difference between the two single phase fluid streams at one end and ΔT_2 is the temperature difference at the other end. This heat exchanger is
- a) A condenser
 - b) An evaporator
 - c) A counter flow heat exchanger
 - d) A parallel flow heat exchanger
66. In counter flow heat exchanger between the two fluids,
- A. Both the fluids at inlet are at the hottest state
 - B. Both the fluids at exit are at the hottest state
 - C. One of the fluids in the hottest state and the other in the coldest state
 - D. Any combination possible depending on the heat exchanger design
67. The rate equation can be found experimentally means
- A. Characterizing the rate of reaction
 - B. Designing the reactor

- C. Finding the order of reaction and rate constant
- D. Measuring variation of concentration with time

68. Which one of the following is the design equation of packed bed reactor

- A. $F_{A,O} dX_A = (-r_A) dV$
- B. $C_{A,O} dX_A = (-r_A) dV$
- C. $F_{A,O} dX_A = (-r_A) dW$
- D. $C_{A,O} dX_A = (-r_A) dW$

69. If the time required to complete a definite fraction of reaction varies inversely as the concentration of the reactants, then the order of reaction is

- A. Negative Zero order
- B. Zero order
- C. Second order
- D. Third order

70. Sizing (/or designing) of a reactor means

- A. Developing the rate of equation
- B. Determining the reactor volume necessary to achieve a specified conversion and rate of production
- C. Finding the appropriate kinetic equation
- D. Expressing the concentration of the reacting species in terms of conversion

71. The design equation of batch reactor is

- A. $t_r = n_{A,O} \int_0^{X_A} \frac{dX_A}{(-r_A)V_f}$
- B. $t_r = -n_{A,O} \int_0^{X_A} \frac{dX_A}{(-r_A)V_f}$
- C. $t_r = -C_{A,O} \int_0^{X_A} \frac{dX_A}{(-r_A)V_f}$
- D. $t_r = C_{A,O} \int_0^{X_A} \frac{dX_A}{(-r_A)V_f}$

72. Which one is true about characteristics of ideal plug flow reactor, PFR?

- A. There is no complete radial mixing of fluid inside the reactor
- B. The density of the flowing system may vary in the direction of flow
- C. The fluid properties are uniform in composition, temperature anywhere in the vessel

- D. The operation of the reactor is unsteady state
73. For a chemical reaction $A \rightarrow B$, it is found that the rate of reaction doubles when the concentration of A is increased four times. Find order of this reaction.
- A. 0.5
 - B. 0.25
 - C. 2
 - D. 1
74. What are the measures of reactors performance based on material equation?
- A. Conversion, Yield, and Rate constant
 - B. Fraction of unreacted, Yield, and Rate constant
 - C. Conversion, Yield, and Rate of reaction
 - D. Conversion, Yield, and Selectivity
75. The fractional volume change of the system for the isothermal gas phase reaction, $A \rightarrow 4B$, between no conversion and complete conversion of reactant A with 50% inert present at the start is
- A. 3
 - B. 2
 - C. 1.5
 - D. 2.5
76. As trainee chemical engineers, if you test the variation of concentration with time in an irreversible batch reaction and you find that the rate equation is independent of the concentration of the participating species, what order of the reaction do this test represent?
- A. First order
 - B. Second order
 - C. Zero order
 - D. Negative order
77. Which one is false for factors that affect the rate of reaction?
- A. The nature of the species involved in the reaction
 - B. Temperature and pressure of a system
 - C. Wave-length of incident radiation
 - D. Type of the system (reactor)
78. The four properties of a system viz. P, V, T, S are related by _____ equation.

- A. Gibbs-Duhem
- B. Gibbs-Helmholtz
- C. Maxwell's
- D. Clayperon

79. Which of the following properties describe entropy?

- A. point function, intensive property
- B. point function, extensive property
- C. path function, extensive property
- D. path function, intensive property

80. Extensive properties of a thermodynamic system depend upon the _____ of the system.

- A. Specific Volume
- B. Temperature
- C. Mass
- D. Pressure

81. Heat of formation of A, B, C and D are 3J, 5J, 4J and 8J respectively. What is the heat of reaction $4A+5B \longrightarrow 2C+2D$?

- A. -10J
- B. -7J
- C. -5J
- D. 0

82. Which one of the following statements regarding chemical equilibria is false?

- A. Catalysts do not alter the position of equilibrium: they do not shift the equilibrium to the left or right
- B. At equilibrium, $G = 0$
- C. If G for a reaction is negative, the forward reaction happens spontaneously
- D. If the equilibrium constant is very large, G is positive

83. Which one of the following statement is **incorrect** regarding agglomeration?

- A. Agglomerates can be obtained using binders or in a binderless manner
- B. In tumble agglomeration fine particles are brought into contact with each other in a flowing system in the presence of binders
- C. in growth agglomeration pressure forces act on a confined mass of particulate solids, which is then shaped and densified
- D. In agglomeration large degree of densification, resulting in low product porosity.

84. Which type of conveying equipment you will recommend for Transporting coal from the mine to the thermal power plant situated at more than 3 km

- A. Screw
- B. Belt
- C. Apron

D. Bucket

85. Is a condition in which the solid particles are arranged randomly forming a non- moveable stationary bed of solids?

A. Fluidized bed

B. Packed bed

C. Fixed bed

D. Movable bed

86. A packed bed of solid particles of density 2500 kg/m^3 , occupies a depth of 1m in a vessel of cross-sectional area 0.04m^2 . The mass of solids in the bed is 59 kg and the surface volume mean diameter of the particles is 1 mm. A liquid of density 800 kg/m^3 and viscosity 0.002 Pa.s flows upwards through the bed. Calculate the voidage.

A. 0.59

B. 0.41

C. 0.0236

D. 0.9764

87. A unit operation in which the insoluble solid component of a solid–liquid suspension is separated from the liquid component by passing the suspension through a porous barrier which retains the solid particles on its upstream surface is known as

A. Sedimentation

B. Clarification

C. Filtration

D. Adsorption

88. Ball mill is used for

A. Course crushing

B. Ultra-fine reduction

C. Intermediate and fine reduction

D. Cutting

89. Which of the following is a characteristic of an adiabatic process?

(A) $\Delta U = 0$

(B) $W = 0$

(C) $Q = 0$

(D) $\Delta V = 0$

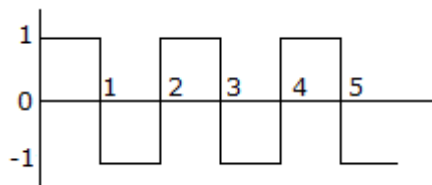
90. Heat does not spontaneously flow from a colder body to a hotter one. Which of the following thermodynamics law states this?

- A. Zeroth law of thermodynamics
- B. First law of thermodynamics
- C. Second law of thermodynamics
- D. Third law of thermodynamics

91. Cascade control means

- A. Feed forward control
- B. More than one feed-back loop
- C. On-off control
- D. One feed-back loop

92. Find the Laplace transform of the following input function shown in figure.



A. $\frac{(1 - e^{-s})}{s(1 - e^{-2s})}$

B. $\frac{2(1 - e^{-s})}{s}$

C. $\frac{(1 - e^{-s})}{s(1 + e^{-s})}$

D. $\frac{(1 - e^{-s}) e^{-s}}{s(1 + e^{-2s})}$

93. The offset introduced by proportional controller with gain K_c in response of first order system can be reduced by

- A. Reducing value of K_c .
 - B. Introducing integral control.
 - C. Introducing derivative control.
 - D. None of the above.
94. Which of the following controllers has maximum offset?
- A. P-controller
 - B. P-I controller
 - C. P-D controller
 - D. D. P-I-D controller
95. Which of the following is an undesirable dynamic characteristic of an instrument?
- A. Reproducibility
 - B. Dead zone
 - C. Time lag
 - D. Static error
96. For a feedback control system to be stable, the
- A. Roots of the characteristic equation should be real
 - B. Poles of the closed loop transfer function should lie in the left half of the complex plane
 - C. Bode plots of the corresponding open loop transfer function should monotonically decrease
 - D. Poles of the closed loop transfer function should lie in the right half of the complex plane
97. In a shell and tube heat exchanger, the outlet temperature of heating/cooling fluid is the _____ variable.
- A. Load
 - B. Manipulated
 - C. Controlled
 - D. Disturbance
98. If response of a control system is to be free of offset and oscillation, the most suitable controller is
- A. Proportional controller
 - B. Proportional-derivative (PD) controller
 - C. Proportional-integral (PI) controller
 - D. Proportional integral-derivative (PID) controller
99. Number of poles in a system with transfer function $1/(s^2 + 2s^2 + 1)$ is

- A. 2
 - B. 5
 - C. 3
 - D. 1
100. When the damping co-efficient (ξ) is unity, the system is
- A. Overdamped
 - B. Critically damped
 - C. Underdamped
 - D. Highly fluctuating

Answers

- 1. A
- 2. C
- 3. D
- 4. B
- 5. D
- 6. A
- 7. B
- 8. D
- 9. B
- 10. C
- 11. A
- 12. A
- 13. B
- 14. A
- 15. C
- 16. B
- 17. B
- 18. C
- 19. C
- 20. B
- 21. B
- 22. B
- 23. B
- 24. B
- 25. D
- 26. C
- 27. B
- 28. B

- 29. A
- 30. B
- 31. C
- 32. D
- 33. D
- 34. D
- 35. D
- 36. C
- 37. B
- 38. A
- 39. C
- 40. D
- 41. C
- 42. C
- 43. C
- 44. A
- 45. B
- 46. C
- 47. B
- 48. D
- 49. C
- 50. C
- 51. C
- 52. C
- 53. C
- 54. B
- 55. D
- 56. B
- 57. C
- 58. C
- 59. A
- 60. A
- 61. C
- 62. C
- 63. A
- 64. A
- 65. C
- 66. A
- 67. C
- 68. C
- 69. C
- 70. B
- 71. A
- 72. B

- 73. A
- 74. D
- 75. C
- 76. C
- 77. D
- 78. C
- 79. B
- 80. C
- 81. B
- 82. D
- 83. C
- 84. B
- 85. C
- 86. A
- 87. C
- 88. C
- 89. C
- 90. C
- 91. B
- 92. A
- 93. B
- 94. A
- 95. C
- 96. B
- 97. C
- 98. D
- 99. C
- 100. B